

LyreFlute: Sensor-Based Musical Experience for COPD Rehabilitation

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Figure 1: Overview of the prototype LyreFlute with embedded sensors and the 3D interactive musical rehabilitation experience.

Abstract

Adherence to at-home COPD breathing exercises remains critically low. We present LyreFlute, a sensor-equipped musical experience that combines accurate breath-tracking with a culturally grounded narrative for respiratory rehabilitation. The system comprises a custom flute with nose and mouth airflow sensors that track breathing patterns, alongside capacitive finger sensors. Users learn songs from an Australian lyrebird through rhythm game mechanics, where correct breathing and finger positioning trigger environmental restoration in a 3D graphics view of the lyrebird's

bushfire-damaged habitat. Our CHI demonstration presents a 3-day condensed musical journey progressing through fire suppression, smoke clearing, and habitat restoration, demonstrating how respiratory rehabilitation can move beyond clinical monotony through meaningful storytelling and at-home physiological monitoring.

CCS Concepts

• **Human-centered computing** → *Ubiquitous and mobile computing*; • **Hardware** → *Sensors and actuators*; • **Applied computing** → *Health informatics*.

Keywords

Breathing, COPD, Rehabilitation, Biofeedback, Flute, Lyrebird

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1 Introduction

Chronic Obstructive Pulmonary Disease (COPD) is a progressive respiratory condition in which the lungs lose their ability to expel air efficiently, leading to breathlessness and reduced exercise capacity [10]. While smoking remains a major risk factor, growing evidence highlights the role of chronic exposure to air pollution as a significant and increasing contributor to COPD, particularly in low- and middle-income countries and regions affected by recurrent bushfires and climate-related environmental change.

Globally, approximately 11% of adults aged 25 and older are affected, and cases are projected to rise from 480 million in 2020 to 592 million by 2050 [3]. In countries such as India, where ambient air pollution levels routinely exceed recommended limits, and in regions experiencing prolonged wildfire seasons, such as Australia, respiratory health burdens are expected to intensify. As these environmental exposures become more widespread, effective, accessible, and scalable rehabilitation strategies are increasingly critical.

At-home breathing exercises such as pursed-lip and diaphragmatic breathing offer accessible daily practices that help patients manage symptoms and improve exercise capacity [18]. However, sustaining engagement and correct technique in unsupervised home-based practice remains challenging with adherence rates of only 16% to 32% across populations [6, 15]. Patients report key barriers as monotonous exercises, difficulty maintaining engagement, and a lack of feedback systems [6]. While digital gamification interventions have emerged as promising solutions to enhance motivation and adherence [2, 8, 13], current applications often fail to provide adequate sensor verification of proper breathing technique or detailed feedback on individual progress.

In this demonstration, we present LyreFlute (see Figure 1), a sensor-equipped musical experience that addresses these challenges through three key innovations. First, we built a custom flute with separate nose and mouth sensors to accurately track breathing patterns, plus finger sensors across four holes to detect hand positioning. Second, we designed a 3D game in which users learn songs from an animated Australian lyrebird through rhythm-based gameplay. The narrative weaves together the 2019–2020 Australian bushfires, which caused widespread respiratory illness [9], with the lyrebird's loss of one-third of its habitat [7]. Third, we built in real-time breath validation that checks whether users maintain proper timing, duration, and intensity before allowing them to progress, while capturing data that could support future clinical monitoring.

Our demonstration invites attendees to experience LyreFlute firsthand, interacting with the sensor flute while immersed in a forest environment guided by a lyrebird. The flute serves as an ideal instrument for breathing rehabilitation, as proper tone production naturally requires the controlled exhalation central to pursed-lip breathing. Participants complete a simulated journey from ecological destruction to restoration, learning how breathing patterns and finger positioning drive environmental change through musical gameplay. Through this hands-on experience, we demonstrate how respiratory rehabilitation can move beyond clinical monotony when culturally resonant stories are combined with precise physiological

monitoring and expressive musical interaction. This contributes to CHI's health informatics and tangible interaction research on embodied physiological interfaces for behavior change.

2 Design

2.1 Design Rationale

The 2019–2020 Australian bushfires provide the context for this project. The fires burned over 18 million hectares, caused widespread respiratory illness, including COPD [9], and destroyed more than a third of the lyrebird's habitat [7]. This allows a narrative where users heal their lungs while restoring the lyrebird's home.

The Australian lyrebird (*Menura novaehollandiae*) became our guide character for its remarkable vocal abilities. Lyrebirds reproduce complex songs and sounds with extraordinary accuracy, often “with a technical competence exceeding what a human flutist could achieve” [11]. Beyond its vocal skill, the lyrebird acts as a living archive, preserving fragments of forest and human history through songs passed across generations, including a distinctive lute-like dialect learned decades earlier [12]. Its conservation, therefore, represents not only ecological recovery but also the forest's acoustic memory, with breathing guided through birdsong linking lung rehabilitation to restoration of a shared ecological and cultural heritage.

We built the interaction around breath as the primary input. Coupling breath control with musical output through a flute-like device creates a direct connection between physical exercise and aesthetic result. Users coordinate nasal inhalation, oral exhalation, and finger positioning: an embodied interaction akin to learning a real instrument while achieving therapeutic breathing patterns.

2.2 Breathing Exercise Rationale

COPD treatment typically involves several weeks of at-home breathing exercises. For CHI Interactivity, LyreFlute is structured as a three-day experience aligned with the conference format, where each day introduces a new song that emphasizes a different breathing pattern. This allows attendees to either experience a single session as a standalone or return over the course of days to observe progression in the respiratory challenge.

In this condensed format, LyreFlute demonstrates two established COPD rehabilitation techniques. First, **Pursed-lip breathing**: inhaling through the nose and exhaling slowly through pursed lips for at least twice the inhalation duration [1, 4]. The prolonged exhalation creates positive back-pressure that prevents premature airway collapse, a critical mechanism for COPD patients experiencing dynamic hyperinflation [14]. Second, **Diaphragmatic breathing**, which emphasizes predominantly abdominal wall movement during inspiration [5], shifting work away from accessory muscles in the neck and shoulders, increasing tidal volume while decreasing respiratory frequency, and improving ventilatory efficiency. Both techniques are recommended in clinical practice [16].

Each day, LyreFlute presents a different song lasting approximately 40–50 seconds, for a total of around 2 minutes per session. Using an airflow sensor to measure peak expiratory flow and breath duration, the system progresses through three stages inspired by interval training strategies in COPD rehabilitation [17].

Day 1 A single continuous phrase of 17 notes over 21 seconds, focusing on sustained breath control and pursed-lip technique.

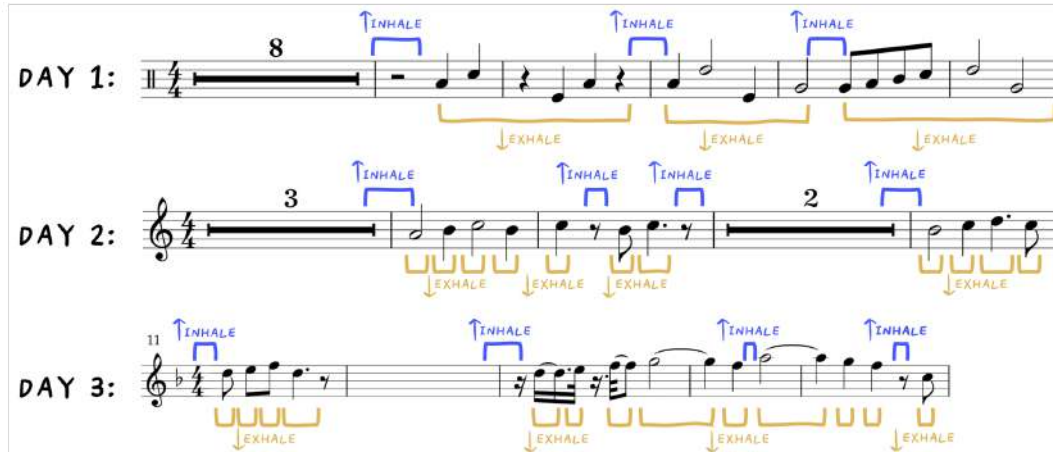


Figure 2: The musical structure of the three-day breathing rehabilitation protocol.

Note durations vary from short grace notes to sustained 4-second tones, training users to modulate exhalation intensity within a single breath cycle;

- Day 2** Three breath groups totaling 35 notes, introducing the inhale-exhale cycling pattern. Users take 5–6 second recovery breaths between groups while playing faster passages (average 0.6s per note). This structure mirrors interval training where brief high-intensity bouts alternate with recovery periods [17];
- Day 3** Four breath groups with 32 notes and progressively shorter 3 to 5 second recovery times between groups. Rest periods between notes decrease to 0.2 seconds, requiring more continuous airflow control. This higher density and reduced recovery create the greatest respiratory demand, combining techniques from previous days.

This three-day progression illustrates key principles of clinical respiratory training, progressing from sustained breath cycles to complex interval patterns that typically unfold over several weeks in rehabilitation settings. An excerpt of the song for each day, along with the corresponding inhalation and exhalation expectations, is shown in Figure 2. While each session functions independently as a flute-playing experience, attendees who return across multiple days can observe the intended progression in breathing challenge.

2.3 The Sensor-Equipped Flute

The LyreFlute device merges traditional flute ergonomics with embedded sensors tracking breathing and fingering. We 3D-printed the housing (322mm length, 32mm diameter). A custom PCB integrates an *XIAO ESP32S3* microcontroller, a *BNO08X* IMU for orientation tracking, and two *FS3000-1015* airflow sensors (0–15 m/sec range, I2C communication at 100–400kHz). A 3.7V, 380 mAh LiPo battery powers the device for about 3 hours. Two *FS3000* sensors separately track nasal inhalation and oral exhalation: one sits inside the flute at its midpoint measuring air velocity during exhalation, while the second mounts on a small ramp atop the flute, angled toward the nose to capture inhalation (see Figure 3). This separation enforces distinct nose-in, mouth-out COPD breathing patterns.

Four finger holes contain 8 mm pure copper rings (5 mm inner diameter) connected to an *MPR121QR2* capacitive touch sensor, detecting on/off states when fingers cover them. With four holes, the system supports 16 possible binary combinations, and we use 12 to map to a one-octave chromatic scale. Unlike physical flutes governed by acoustic physics, we explicitly mapped each finger combination to notes from C4 to G5, using the thumb hole (H4) and three front holes (H1–H3) to encode a 4-bit binary representation. This deterministic mapping enables versatile musical patterns while remaining accessible to non-musicians. The *ESP32S3* processes sensor data in real-time and transmits it via Bluetooth to a desktop/laptop running the game. The desktop system compares breath timing and duration against target patterns for each song section, scoring accuracy, and providing immediate feedback.

2.4 Game Experience Design

LyreFlute is experienced as an embodied breathing interaction within a responsive 3D forest. Users face a display showing an environment that responds directly to their breathing, linking respiratory exercise with ecological healing. Built in Unity, the forest evolves across three days to match the narrative arc: Day 1 shows burning landscapes with flames and smoke, Day 2 ash covered ground with lingering embers, and Day 3 restored ecosystems with vegetation and wildlife, as shown in Figure 4.

Correct breathing exercises trigger immediate environmental responses: rain douses flames, wind clears smoke and ash, and plants grow from scorched earth. An animated lyrebird perched on a tree trunk guides users through exercises in call-and-response patterns, demonstrating songs they replicate with their sensor flute.

We adapted rhythm game mechanics for breathing rehabilitation. Musical notes scroll down four vertical tracks matching the flute’s finger holes, as shown in Figure 5. To play successfully, users coordinate nasal inhalation timing, finger positioning, and oral exhalation with proper intensity and duration. Correct notes produce sound and trigger environmental effects.

Audio processing uses a custom .NET application that allows Unity to host VST plugins directly without an external DAW. Unity



Figure 3: The LyreFlute hardware design with embedded airflow sensing configuration.



Figure 4: 3D graphics showing the narrative-driven forest environments across the rehabilitation journey.

receives ESP32 data, processes it, and converts it into MIDI notes for a VST host running Monstera Ethnica with Suling Sunda flute presets. User performance shapes the environment: Day 1 rainfall follows accuracy, Day 2 wind reflects breath intensity to disperse smoke, and Day 3 introduces duets where successful exchanges bring flowers and trees, culminating in full forest restoration.

3 Interactivity Implementation

For the CHI demonstration, we condensed the standard two week COPD breathing rehabilitation into a three day format, with sessions lasting about two minutes. The demo station includes the sensor flute with individually wrapped disposable mouthpieces and external speakers for bystanders. The flute is sanitized between users, and an attendee assists with setup and explains the basic mechanics. We selected representative songs to showcase the system within the time constraints. Day 1 features a simplified fire extinguishing song focused on sustained exhalations. Day 2 introduces alternating breath patterns to demonstrate smoke clearing. Day 3 presents a short duet with the lyrebird that combines all techniques. Participants progress through all three days regardless of performance so everyone experiences the full narrative without creating queues. External audio and a monitor allow observers to hear the music and see coordination between breath and finger movements as players move through the forest restoration sequence. After each

day, participants view a summary screen showing accuracy as a percentage of environmental restoration achieved.

4 Future Work

LyreFlute shows how to gamify COPD rehabilitation, but there are many ways it could be extended. Our immediate next steps are technical enhancements and clinical validation. We plan to implement octave changing controlled by exhalation velocity, expanding the musical range and challenging progression. To increase accessibility, we will develop a low-cost Android integration. Clinical trials will evaluate LyreFlute with a COPD cohort over 15 days, measuring physiological effects through comprehensive monitoring, including full breathing straps and ECG, while tracking completion rates and adherence patterns, compared to traditional training methods.

In the longer term, we plan to explore culturally significant flutes from different regions that incorporate alternative breathing techniques relevant to rehabilitation. This includes instruments like the kōauau and bamboo flute, selected to resonate with the cultural backgrounds of specific COPD patient cohorts. By adapting both the physical instrument and the narrative context to match local communities, we aim to maintain the core principle of connecting personal healing with meaningful cultural stories while ensuring respiratory exercises remain clinically effective.

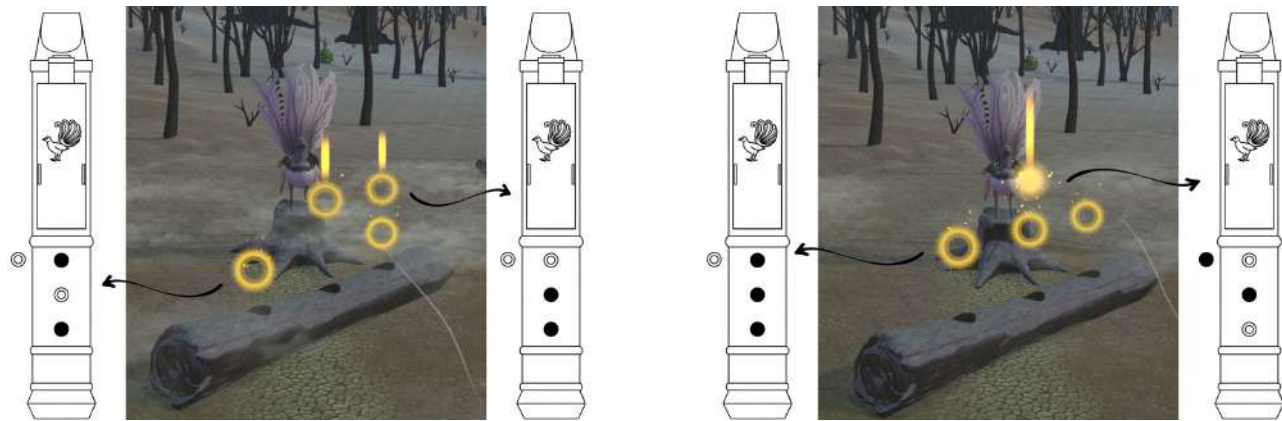


Figure 5: The rhythm game interface linking breathing and finger input to environmental feedback.

5 Conclusion

We presented LyreFlute, a sensor-equipped musical experience that transforms COPD breathing exercises into an engaging narrative connecting personal lung healing with lyrebird habitat restoration. By combining accurate breath tracking via airflow sensors, rhythm game mechanics, and culturally grounded storytelling, respiratory rehabilitation can move beyond clinical monotony to incorporate aesthetic experience and emotional engagement. This work prompts reflection on whether embedding health interventions within personally meaningful narratives can transform adherence from clinical compliance into genuine engagement. Future clinical trials will validate whether this approach can effectively overcome the adherence barriers of traditional at-home breathing programs.

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